



TN CHENNAI DISTRICT STATEBOARD QUESTION PAPER 2024 - 2025

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Class : 11

REVISION EXAMINATION - 2025

Time Allowed : 3.00 Hours]

PHYSICS

[Max. Marks : 70

PART-I

Note: (i) Answer all the questions

15x1=15

(ii) Choose the most suitable answer from the given four alternatives and write the option code and the corresponding answer.

- If the mass of the Earth and Sun suddenly double, the gravitational force between them will
(a) remain the same (b) increase 2 times (c) increase 4 times (d) decrease 2 times
- With an increase in temperature the viscosity of liquid and gas respectively will
(a) Increase and Decrease (b) Increase and Decrease
(c) Decrease and Increase (d) Decrease and Decrease
- The dimensional formula of Planck's constant h is
(a) $[ML^2T^{-1}]$ (b) $[ML^2T^{-3}]$ (c) $[MLT^{-1}]$ (d) $[ML^3T^{-3}]$
- Identify the unit vector in the following.
(a) $\hat{i} + \hat{j}$ (b) $\frac{\hat{i}}{\sqrt{2}}$ (c) $k - \frac{\hat{j}}{\sqrt{2}}$ (d) $\frac{\hat{i} + \hat{j}}{\sqrt{2}}$
- The angle between the two vectors $\vec{A} = 2\hat{i} + 4\hat{j} + 5\hat{k}$ and $\vec{B} = \hat{i} + 3\hat{j} + 6\hat{k}$ is
(a) 15° (b) 30° (c) 45° (d) 60°
- 1 kilo watt hour is equal to
(a) 3600 J (b) $36000 \times 10^3 J$ (c) 3.6KJ (d) $3.6 \times 10^5 J$
- When a car takes a sudden left hand turn in the curved road, passengers are pushed towards the right due to
(a) Inertia of direction (b) Inertia of motion (c) Inertia of Rest (d) Absence of Inertia
- A body of mass 1 kg is thrown upwards with a velocity 20 ms^{-1} . It momentarily comes to rest after attaining a height of 18 m how much energy is lost due to air friction ?
(a) 20 J (b) 30 J (c) 40 J (d) 10 J
- The speed of the centre of a wheel rolling on a horizontal surface is V_0 . A point on the rim in level with the centre will be moving at a speed of
(a) Zero (b) V_0 (c) $\sqrt{2}V_0$ (d) $2V_0$
- When a uniform rod is heated which of the following quantity of the rod will increase
(a) Mass (b) Weight (c) Centre of mass (d) Moment of inertia
- If the Temperature and pressure of a gas, is doubled the mean free path of the gas molecules?
(a) Remains same (b) Doubled (c) Tripled (d) quadrupled
- A particle executing SHM crossed points A and B with the same velocity. Having taken 3 s in passing from A to B, It returns to B after another 3s. The time period is :
(a) 15 s (b) 6s (c) 12 s (d) 9 s
- Equation of travelling wave on a stretched string of linear density 5 g / m is $y = 0.03 \sin (450 t - 9x)$ where distance and time are measured in SI units. The tension in the string is :
(a) 5 N (b) 12.5 N (c) 10 N (d) 7.5 N
- The Value of Stefan's constant is
(a) $5.67 \times 10^8 \text{ W m}^{-2} \text{ K}^{-4}$ (b) $5.67 \times 10^{-8} \text{ W m}^{-2} \text{ K}^{-2}$
(c) $5.67 \times 10^{-8} \text{ W m}^{-2} \text{ K}^{-4}$ (d) $5.67 \times 10^{-8} \text{ W m}^{-2} \text{ K}^{-4}$
- The Degree of Freedom for diatomic molecule at high temperature is
(a) 3 (b) 5 (c) 6 (d) 7



PART - II

II. Note : Answer any six questions. Question Number 24 is Compulsory.

6x2=12

16. Write down the Kinematic equations for Angular motion.
17. What are Inertial Frames?
18. Calculate the work done by a force of 30 N in lifting a load of 2kg to a height of 10m. ($g = 10 \text{ ms}^{-2}$)
19. What is radius of Gyration?
20. What are Geostationary satellite?
21. Two pistons of a hydraulic lift have diameters of 60cm and 5cm. what is the force exerted by the larger piston when 50N is placed on the smaller piston?
22. Write down the factors affecting velocity of Sound in gas.
23. Define forced oscillation, Give an example.
24. During a cyclic process, a heat engine absorbs 500 J of heat from a hot reservoir, does work and ejects an amount of heat 300 J into the surroundings (cold reservoir). Calculate the efficiency of the heat engine?

PART - III

III. Note : Answer any six questions. Question Number 33 is Compulsory.

6x3=18

25. Write down the Characteristics of Stationary Waves.
26. Compare single harmonic motion and angular harmonic motion.
27. Ten particles are moving at the speed of 2, 3, 4, 5, 5, 5, 6, 6, 7 and 9 ms^{-1} . Calculate the rms speed, average speed and most probable speed.
28. Derive Poiseuille's formula for the volume of a liquid flowing per second through a pipe under streamlined flow.
29. State Kepler's three laws.
30. What are the limitations of dimensional analysis?
31. Two vectors are given as $\vec{r} = 2\hat{i} + 3\hat{j} + 5\hat{k}$ and $\vec{F} = 3\hat{i} - 2\hat{j} + 4\hat{k}$. Find the resultant vector. (Torque)
32. Explain the need for banking of tracks.
33. A fly wheel rotates with a uniform angular acceleration. if its angular velocity increases from $20\pi \text{ rad / s}$ to $40\pi \text{ rad / s}$ in 10 second. Find the number of rotations in that period.

PART - IV

IV. Note : Answer all the questions.

5x5=25

34. (a) Obtain an expression for the time period T of a simple pendulum. The time period T depends on (i) mass of the bob (m) (ii) length of the pendulum (l) and (iii) acceleration due to gravity (g) at the place where the pendulum is suspended. (Constant $K = 2\pi$) (OR)
- (b) Describe Newton's formula for velocity of sound waves in air and also discuss the Laplace's correction.
35. (a) Discuss in detail the energy in simple harmonic motion. (OR)
- (b) Derive the expression for moment of inertia of a uniform ring about an axis passing through the centre and perpendicular to the plane.
36. (a) State and explain Work energy principle mention any three examples for it. (OR)
- (b) Derive Mayer's relation for an ideal gas.
37. (a) Explain the different types of Modulus of elasticity. (OR)
- (b) Derive the kinematic equations of motion for constant acceleration.
38. (a) Briefly explain two origin of Friction. show that in an inclined plane, angle of friction is equal to angle of repose. (OR)
- (b) Explain the variation of 'g' with altitude.